DAVE User guide

DAVE is an AI-powered framework designed for assisted analysis of document collections in knowledge-intensive domains. It enables domain experts to efficiently explore and analyze large text corpora through a combination of:

- **Entity-driven faceted search** for structured information retrieval
- **Conversational AI interface** for natural language queries
- **Interactive entity annotation and correction** for improved knowledge management

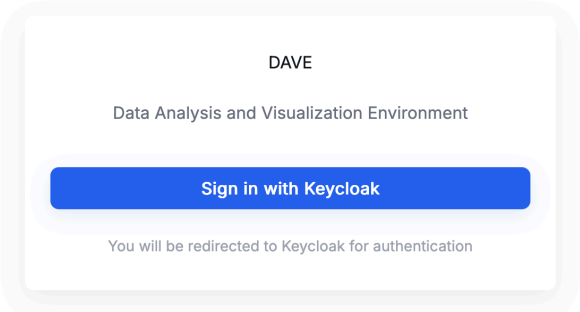
DAVE is particularly useful in domains such as **law, healthcare, finance, and real estate**, where factual data is closely tied to entities and their relationships.

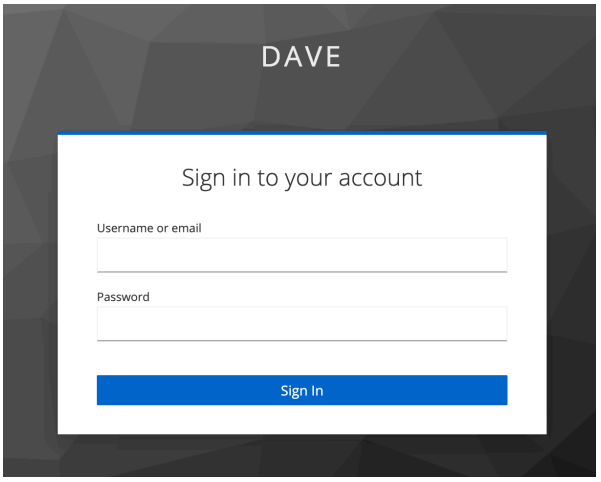
In this user guide we will detail the following views:

- Login
- Homepage
- Faceted Search
- Advance Browsing
- QA Assistant
- Collection Management
- Settings
- Annotation Pipeline Configuration

Login

In DAVE, users can authenticate using their Keycloak credentials by clicking the **Sign in with Keycloak** button. This action redirects the user to the Keycloak login page. Upon successful authentication, the user is redirected back to DAVE's home page.

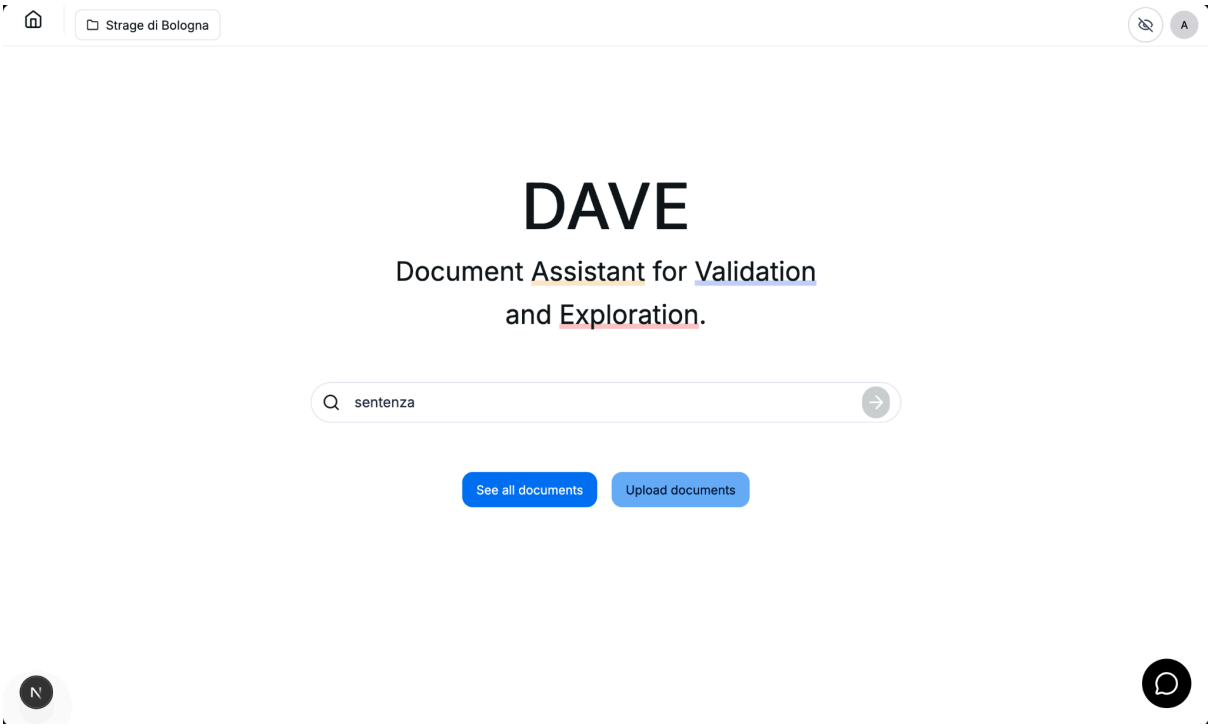


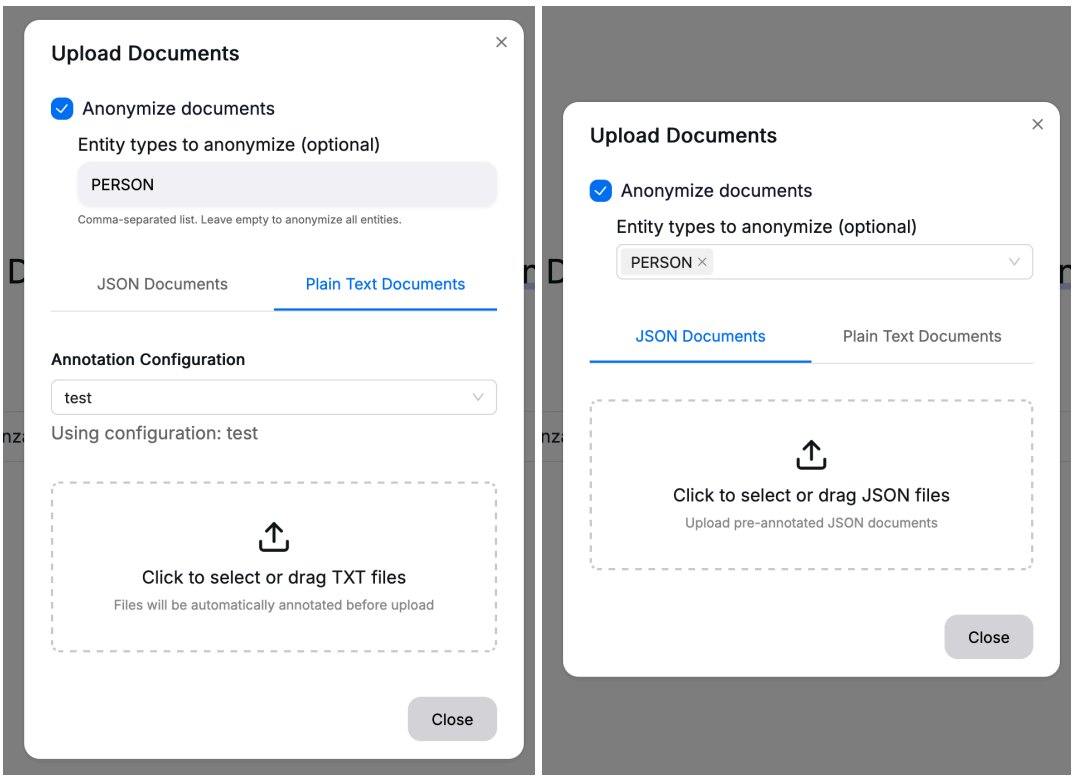


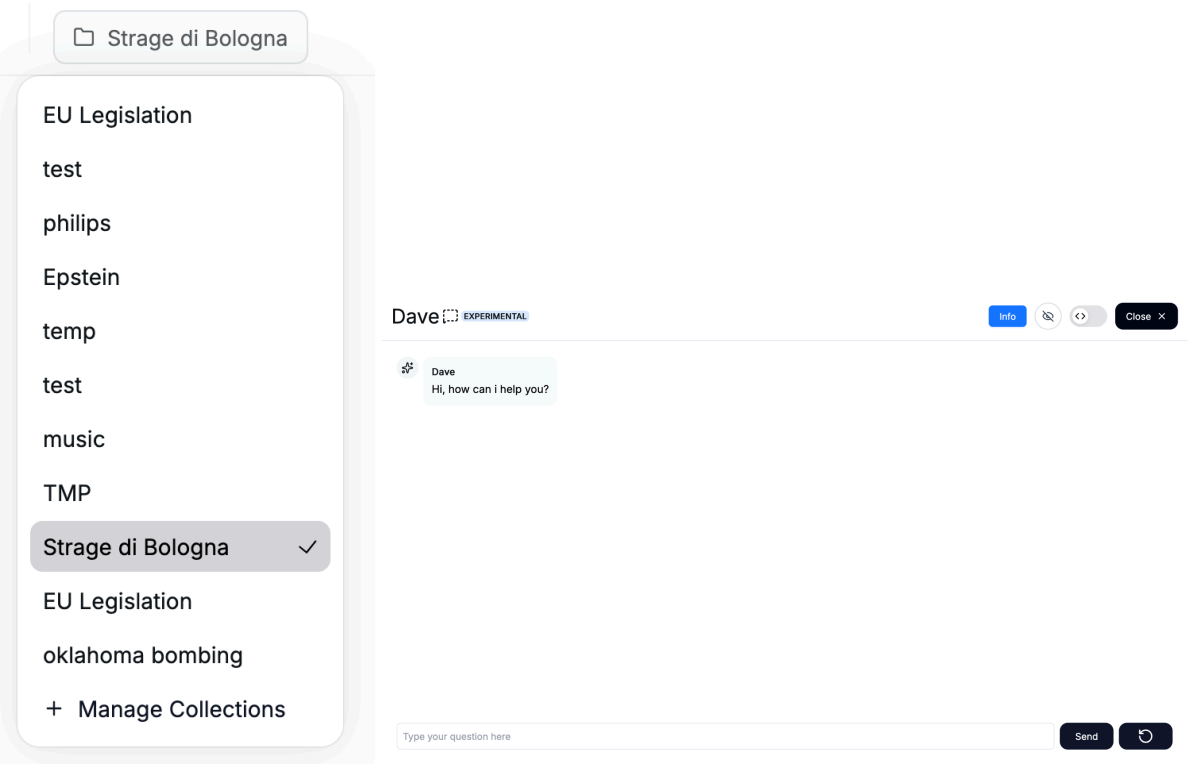
Homepage

From the DAVE homepage, users can perform the following actions within the currently active document collection:

- Search for documents
- View all documents in the collection
- Switch to a different document collection
- Ask questions using the chat interface
- Upload new documents to the current collection



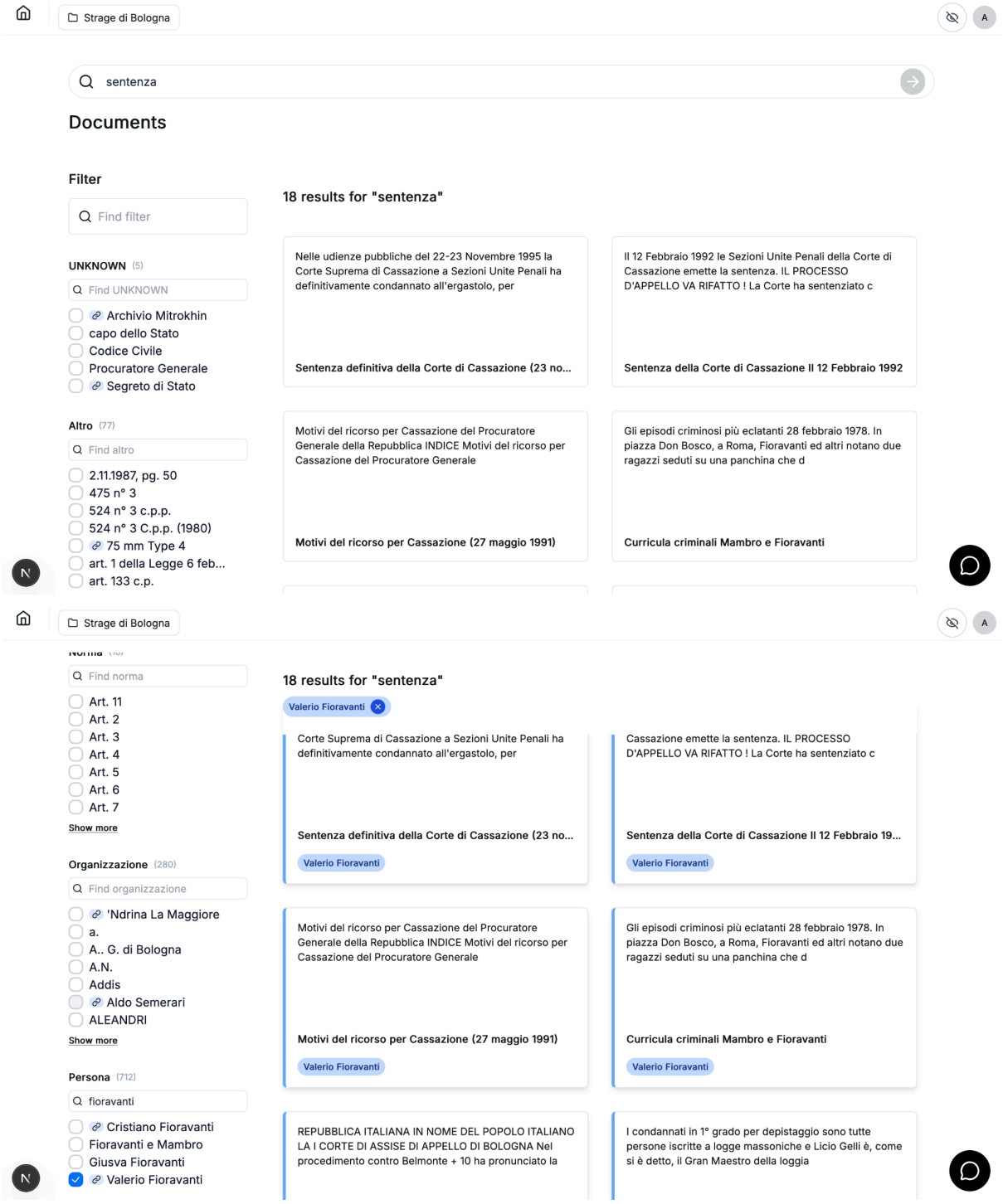




Faceted search

The faceted search feature enables users to perform a full-text search across all documents in the currently active collection. Additionally, users can refine search results by applying semantic filters, which are generated from entities identified within the document corpus.

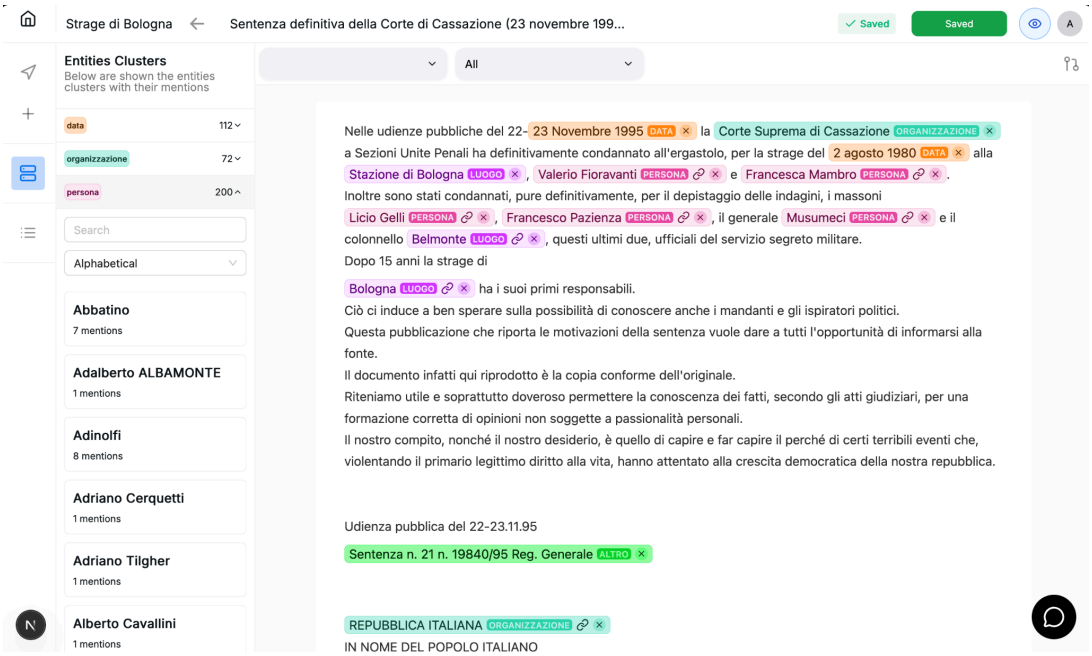
Using a functionality built into the chat module, users can ask questions restricted to the documents returned by the full-text search and the applied filters.

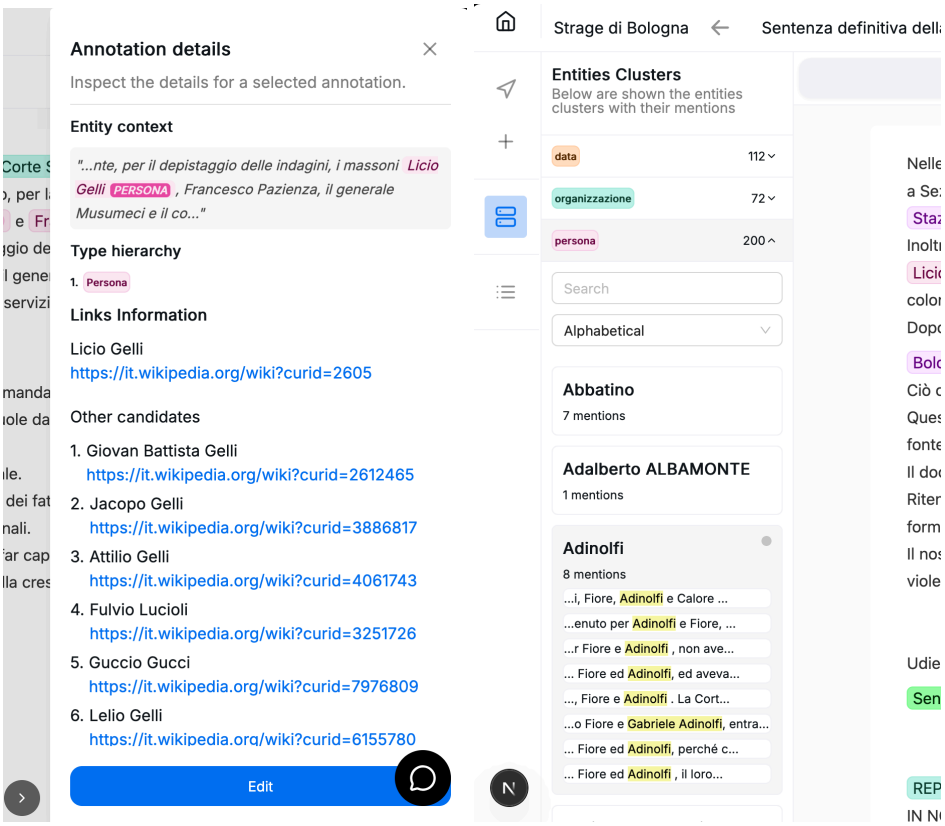


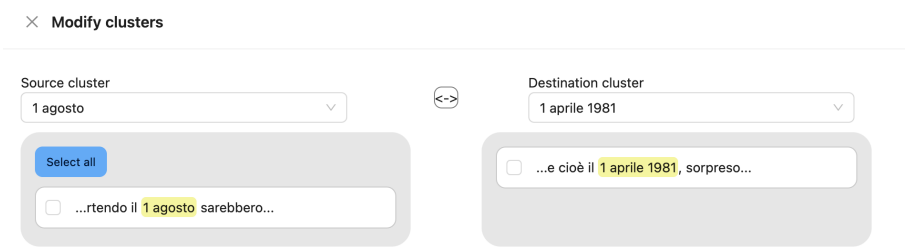
Advance Browsing

Each document in a collection can also be explored individually. When users click on a document from the faceted search results, they are taken to the document page. This page displays the document's annotated text, including all entity mentions. The page also includes a cluster list representing the entities found in the document. For each entity, users can view the list of mentions and, by clicking on them, see those mentions highlighted within the

document. Additionally, users can click on any entity mention directly in the document text to view detailed information, including entity type and linking data. Users can also manually annotate mentions using the “+” tool in the left toolbar, or delete mentions by pressing the “x” button in the highlighted entity mention. Furthermore, by clicking the **Edit clusters** button, users can drag and drop entity mentions from one source cluster to a destination cluster to correct errors in cluster detection. Also, if enabled, users can ask questions in the chat using the current document as the single source.





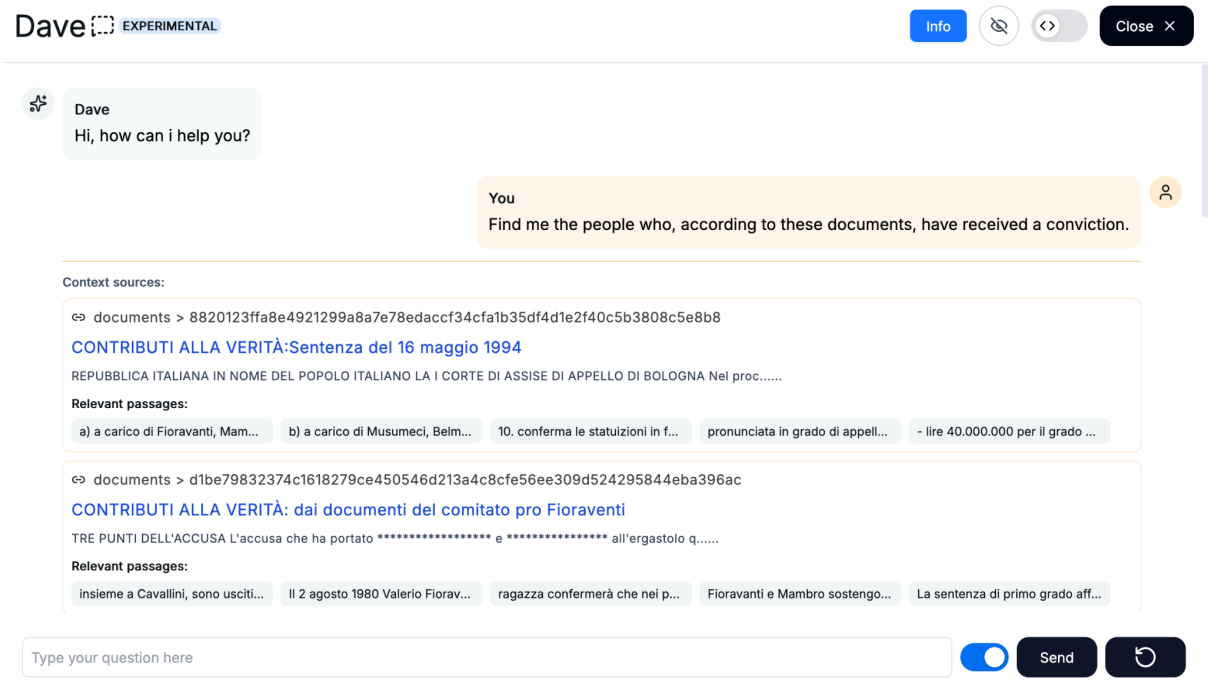


QA Assistant

This module enables users to ask questions or perform conversational search using the documents in the current collection as the source. The assistant answers display all document chunks used for response generation, grouped by document provenance, along with the generated response and citations linking to the source chunks.

Users can toggle the switch located to the right of the send button to restrict the source documents to either the current faceted search results or the currently opened document. When left disabled, all documents in the collection are considered as source documents.

Additionally, a **DEV Mode** is available, allowing users to adjust and fine-tune text generation parameters, including temperature, maximum length, system prompt, retrieval method, and other advanced settings.



Dave EXPERIMENTAL

Info Close ×

Dave

Based on the provided documents, the following people are explicitly mentioned as having received a conviction (condannati):

[4]

Cavallini and **Giuliani**: They were condemned for the crime of armed band (*banda armata*) in an appeal court judgment (*grado di appello*).

[4]

Musumeci and **Belmonte**: They were condemned in Rome with a final judgment (*sentenza definitiva*) for peculation (*peculato*).

[10]

They were also listed as defendants in Bologna for the diversion (*depistaggio*) of the information regarding the Bologna massacre.

[21]

Fioravanti, **Mambro**, **Picciafuoco**, and **Gelli** (mentioned in the context of being imputati/imputati): While the text mentions fines imposed on them (*a carico di*), it does not explicitly state they received a criminal conviction in the same definitive manner as Cavallini, Giuliani, Musumeci, or Belmonte, though they are described as "imputati" (defendants) in the judgment.

[4]

Specifically confirmed convictions mentioned:

Cavallini and **Giuliani** (condannati per il delitto di banda armata).

[4]

Type your question here

Send

You are currently in development mode

Info Close ×

owing people are explicitly
(condannati):

ned for the crime of armed band
nt (*grado di appello*).

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massacre.

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criminal conviction in the same
sumeci, or Belmonte, though they
the judgment.

ned:
litto di banda armata).

oma con sentenza definitiva per

Send

Temperature 0.7

Max new tokens 1024

Top p 0.65

Top k 40

Frequency penalty 1.15

Retrieval method

Hybrid Retrieval Hibrid Retrieval No NER
Dense Full text None

Force RAG

Use Multi-Agent System

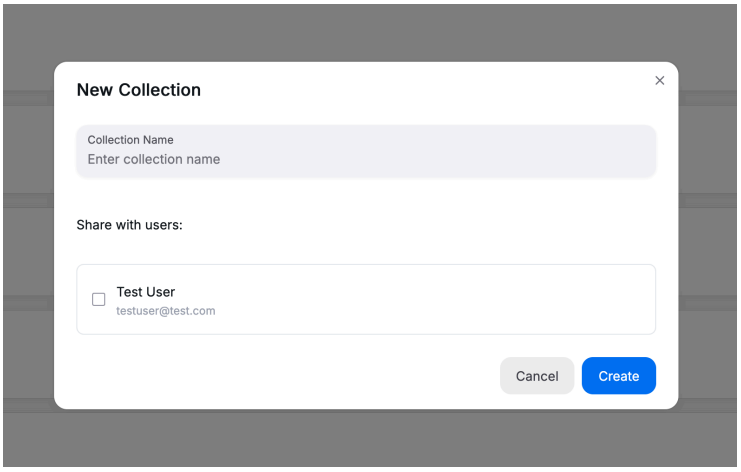
System prompt

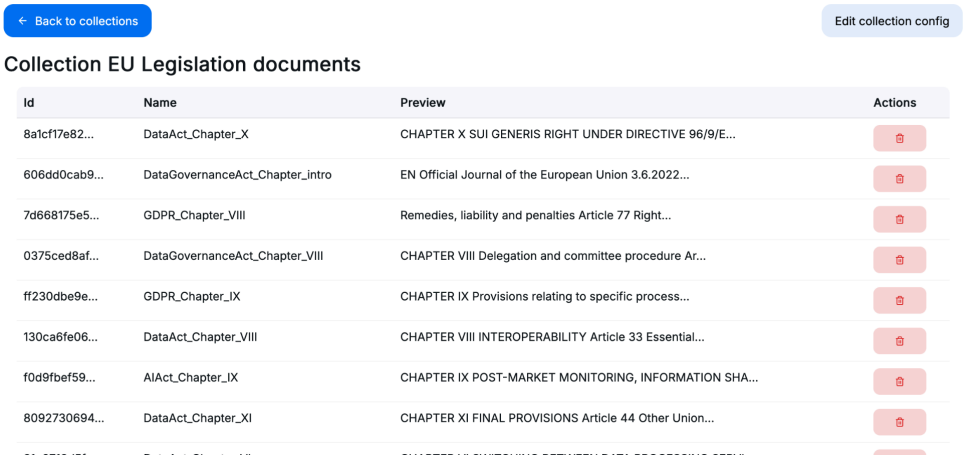
You are an expert assistant that answers questions based on provided context.

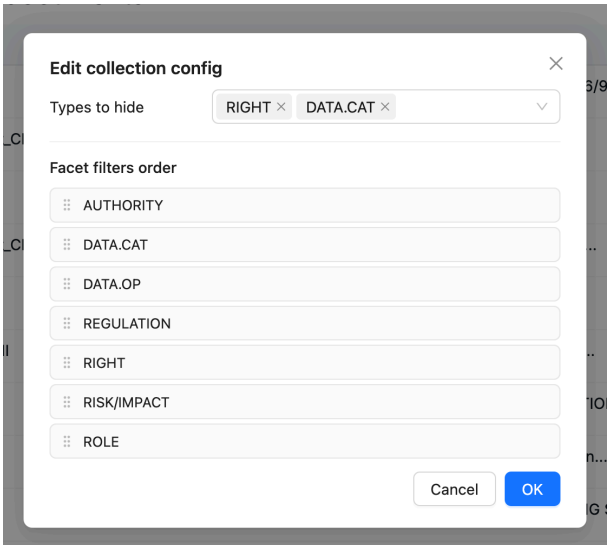
Collection Management

Users can create and manage document collections. Each collection has a name and a list of users who can access it, in addition to the owner. Users can create, rename, or delete collections. Deleting a collection will also delete all documents contained within it.

Each collection includes a dedicated page where users can view the document list, upload new documents, and modify collection settings. The collection settings panel allows users to specify which entity types should be hidden from the faceted search. Additionally, users can change the display order of faceted search filters by dragging and dropping the entity types.

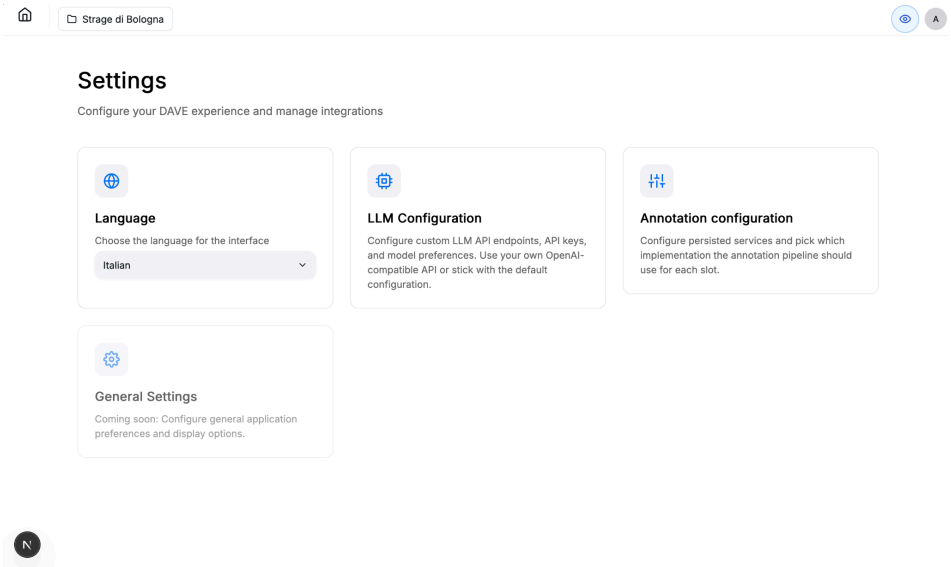






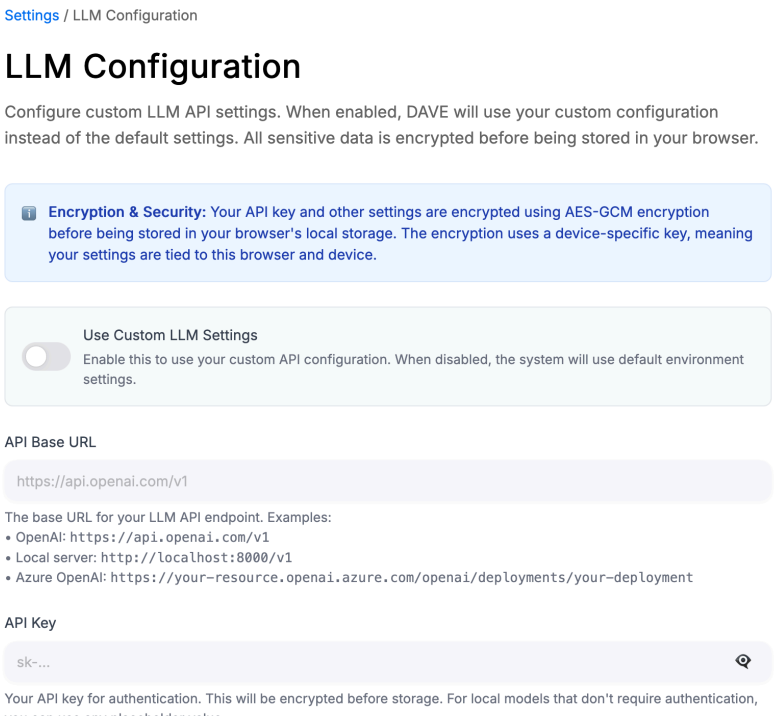
Settings

From the settings page, users can modify the default parameters for the chat module. Additionally, users can configure an annotation pipeline, which can be selected when uploading a TXT document to be annotated.



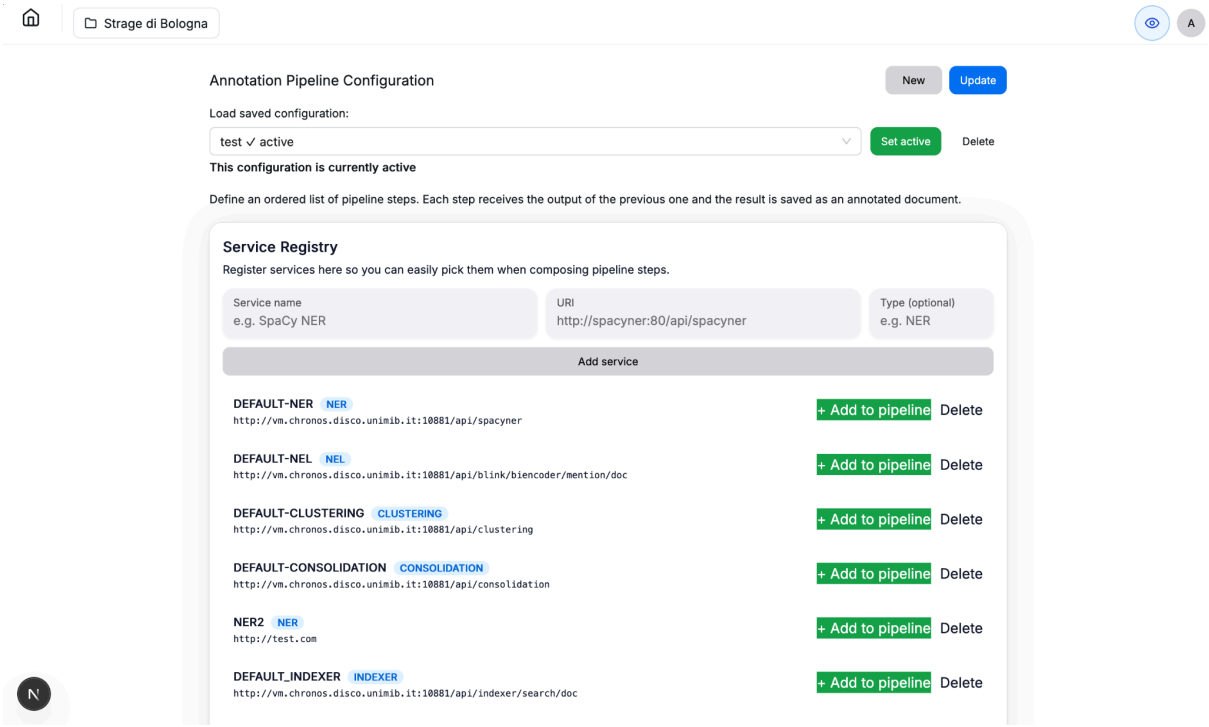
QA Assistant settings

From the settings page, users can configure a custom LLM provider by adding an OpenAI-compatible endpoint, API key, and model name. Users can also modify the default text generation parameters, including temperature, system prompt, and other related settings.



Annotation Pipeline configuration

From this page, users can create different pipelines to be used during the upload of TXT documents. Users can add individual service endpoints (e.g., NER, NEL, nil prediction) and then create a pipeline by arranging the services in the desired order. During the annotation process, DAVE will call these services sequentially, passing the output of each annotation phase to the next one. Users can also create a pipeline consisting of a single service.



Strage di Bologna

A

Pipeline Steps

Steps are executed in order. The document is passed from one step to the next.

+ Add step

1. **DEFAULT-NER** NER
   `http://vm.chronos.disco.unimib.it:10881/api/spacyner`

2. **DEFAULT-NEL** NEL
   `http://vm.chronos.disco.unimib.it:10881/api/blink/biencoder/mention/doc`

3. **DEFAULT_INDEXER** INDEXER
   `http://vm.chronos.disco.unimib.it:10881/api/indexer/search/doc`

Preview


```
[
  {
    "id": "6954f44fd92737d942e691ba",
    "name": "DEFAULT-NER",
    "uri": "http://vm.chronos.disco.unimib.it:10881/api/spacyner",
    "serviceType": "NER"
  },
  {
    "id": "6954f44fd92737d942e691c4",
    "name": "DEFAULT-NEL",
    "uri": "http://vm.chronos.disco.unimib.it:10881/api/blink/biencoder/mention/doc",
    "serviceType": "NEL"
  },
  {
    "id": "697738467dbba8de24f5fc83",
    "name": "DEFAULT_INDEXER",
    "uri": "http://vm.chronos.disco.unimib.it:10881/api/indexer/search/doc",
    "serviceType": "INDEXER"
  }
]
```